SINKGRAB · TRACEABILITY MATRIX

SinkGrab traceability matrix

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/sinkgrab		
Source	docs/03b-traceability.md @ ec186d3		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-10-25	Amish Chadha	Generated by .kit/trace.py
0.3	2026-10-25	Amish Chadha	Generated by .kit/trace.py
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SinkGrab traceability matrix

Generated by `python .kit/trace.py` from the user requirements, product requirements, calculations, build plan and BOM. Do not edit by hand. 13 user requirements, 13 product requirements, 0 gaps.

Table 1. User requirement to product requirement to evidence to verification.

User requirement	Product requirement	Title	Evidence	Verification	Status
UR-001	PR-001	Depth below the water table	SKG-CAL-001, SKG-BLD-001, SKG-STM-001	Field trial in a partner well, dip tape before and after (TRL 5)	The 8-ring string sinks only while skin friction stays below 1.66 kPa; the central estimate for disturbed sand (2 kPa) exceeds it and soil in full contact gives about 25 kPa (STEM S21). Unaided this is **not met on paper**; with the bentonite slurry in the overcut annulus that Amish chose on 2026-10-25 (SKG-DEC-001 O2) it is **at risk** until the test pit shows the lubricated friction below the limit
UR-004	PR-002	Ring sizes	SKG-CAL-001, SKG-BLD-001	Fit check in rings of three sizes (TRL 4)	Met by design: grab 468 mm across; scraper folds to 387 mm radius on 300 mm of sleeve travel (arm pivots at 60 mm, SKG-DEC-001 K6); toes, skids and a longer arm pair cover the range
UR-005	PR-003	Bite	SKG-CAL-001, SKG-BLD-001, SKG-STM-001	Test-pit trial with a measured tub (TRL 4)	At risk:** 21.2 L at 75 % fill, 3:1 tackle; 14.1 L at 50 %; closing margin over passive drag between about 0.9 and 1.8 once the empty grab and sheave friction are counted (STEM S2)
UR-005, UR-006	PR-004	Cycle time	SKG-CAL-001, SKG-BLD-001, SKG-STM-001, SKG-OPT-001	Timed trial over 20 cycles (TRL 4)	Not met on paper by about 0.2 min:** 3.19 min with three at the hoist (3.8 min with two); the timed trial decides
UR-006	PR-005	Crank force	SKG-CAL-001, SKG-BLD-001, SKG-STM-001	Spring balance on the crank handle (TRL 4)	Met on paper: 55 N each with two people (110 N for one)
UR-003	PR-006	Tilt	SKG-CAL-001, SKG-BLD-001	Four-point dip tape readings every cycle in trials (TRL 5)	To verify at TRL 4 and 5: readings resolve about 1 in 250; correction by scraping the high side and saddles
UR-007	PR-007	Proof load	SKG-CAL-001, SKG-BLD-001, SKG-STM-001	CalRig proof-load test with record, clutch slip test on each rope layer, load indicator calibration (TRL 4)	At risk:** proof 188 kg, inside the HatchSide proof of 225 kg; the clutch slips at 1,127 to 1,472 N against a steady full-grab pull of 1,068 N (margin 1.06 on the third layer) on a ±5 % setting that only

User requirement	Product requirement	Title	Evidence	Verification	Status
					CalRig can show (STEM S4, S6, S7, E2; SKG-CAL-001 C, G4)
UR-002, UR-013	PR-008	Nobody in the well	SKG-CAL-001, SKG-STM-001	Trial log and method statement review (TRL 4)	Met by design: every task from the surface; doors cover the shaft
UR-008	PR-009	Portability	SKG-CAL-001, SKG-BLD-001, SKG-STM-001	Weigh each piece; load trial (TRL 4)	Met on paper: heaviest piece 22.4 kg (capstan frame) with the scraper arms carried unpinned (scraper head 27.3 kg with them on, 21.9 kg without); kit about 467 kg, longest piece 2.0 m; the assembled grab (52.7 kg) is moved only on the line (SKG-DEC-001 K3)
UR-009	PR-010	Local build	SKG-CAL-001	Build by a partner workshop from the published drawings (TRL 4)	Met by design; the shell skins need plate rolls, which most fabricators have
UR-010	PR-011	Prototype cost	SKG-CAL-001	Bill of materials (analysis, TRL 3); receipts (TRL 4)	Met on paper, within the target: USD 3,063 (USD 937 under)
UR-012	PR-012	Air test	SKG-BLD-001, SKG-STM-001	BOM and method review (inspection, TRL 3); bump test before use (TRL 4)	Met by design: BOM lines 31 and 35, build plan safety stop 1 (accepted by Amish on 2026-10-25, SKG-DEC-001 O3)
UR-011	PR-013	Soils	SKG-OPT-001	Test-pit trials in each soil (TRL 4)	To verify at TRL 4: closing force on paper in loose sand only (STEM S2); accepted by Amish on 2026-10-25 (SKG-DEC-001 O3)

Gaps

None.